MTH355

Timed Online Assignment – January Semester 2022

Basic Mathematical Optimization

Tuesday, 17 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Explain why we may not always find an LU decomposition for a general square matrix A.

(3 marks)

(b) Consider the following system of linear equations

$$\begin{aligned} 2x_1 + 3x_2 + x_3 &= 4 \\ -6x_1 + 3x_2 + 6x_3 &= 22 \\ 9x_1 - 6x_2 + 6x_3 &= -5 \end{aligned}$$

(i) Write down the matrix A and vector b, where $Ax = b$. Apply the LU decomposition technique on A. Find the elementary matrices E_{ij}.

(9 marks)

(ii) Based on the LU of A in (b)(i), solve the system of linear equations.

(3 marks)

(iii) Use the LU of A to find the first column of A^{-1}.

(3 marks)

(c) Consider the system of linear equations.

$$\begin{aligned} 2x_1 + 6x_2 + x_3 &= 4 \\ -2x_1 + 3x_2 + 6x_3 &= 22 \\ 9x_1 - 6x_2 + 2x_3 &= -5 \end{aligned}$$

Apply the conditions of convergence for a set of linear equations. Starting from $(x_1, x_2, x_3) = (1,2,1)$, perform three iterations of the Gauss-Seidel iterative scheme. Keep all values up to five decimal places. Provide enough details for the iterative process.

(7 marks)

Question 2

(a) Consider the following linear programming (LP) problem.

$$\begin{gathered} \text{Maximize } z = 3x_1 + 7x_2 \\ 5x_1 + 2x_2 \geq 10 \\ 2x_1 + 3x_2 = 6 \\ x_1,\ x_2 \geq 0 \end{gathered}$$

(i) Analyse the LP problem using the two-phase simplex method. Find the optimal solution.

(13 marks)

(ii) Determine the dual model.

(4 marks)

(iii) Use the optimal solution of the primal model and the complementary slackness optimality conditions to compute the optimal solution of the dual model.

(4 marks)

(b) Compare the 2-phase simplex method and the dual simplex method. Explain which method is more robust in handling a general LP problem.

(4 marks)

Question 3

Consider the following linear programming (LP) problem.

$$\text{Maximize } z = -x_1 + 3x_2 + 6x_3$$
$$-x_1 + x_2 + 3x_3 \leq 20$$
$$3x_1 + 2x_2 + 5x_3 \leq 45$$
$$x_1, x_2, x_3 \geq 0$$

We introduce two slack variables x_4 and x_5 for the respective constraints. The optimal tableau is given in Table Q3 below.

Basic	$\boldsymbol{x_1}$	$\boldsymbol{x_2}$	$\boldsymbol{x_3}$	$\boldsymbol{x_4}$	$\boldsymbol{x_5}$	**Solution**
$\boldsymbol{z}$	0	0	$-\frac{13}{5}$	$-\frac{11}{5}$	$-\frac{2}{5}$	-62
$\boldsymbol{x_2}$	0	1	$\frac{14}{5}$	$\frac{3}{5}$	$\frac{1}{5}$	21
$\boldsymbol{x_1}$	1	0	$-\frac{1}{5}$	$-\frac{2}{5}$	$\frac{1}{5}$	1

Table Q3

We implement sensitivity analysis for each of the following changes with regard to the original model.

(a) When the value of b_1 changes from 20 to 25, find the modified tableau. Subsequently solve the new LP problem.

(7 marks)

(b) We allow c_2, the coefficient of x_2 in the objective function to change from 3 to β. Find a range of values for β in which the current basis matrix remains optimal.

(6 marks)

(c) For the values of β in the range found in (b), compute the new maximal value of z. What are the values of the basic variables?

(3 marks)

(d) The coefficient vector of x_3, (c_3, a_{13}, a_{23}), is changed from $(6, 3, 5)$ to (7,2,4). Find the new optimal solution.

(6 marks)

(e) Explain why we would want to reoptimize a modified tableau rather than to solve the LP problem from scratch when changes are made to the constraints.

(3 marks)

Question 4

(a) A car manufacturing company is able to produce five models, M1, M2, …, and M5, in the next 3-year period. The amount, in the company's financial unit of investment to produce each model is shown in Table Q4(a).

Model	**M1**	**M2**	**M3**	**M4**	**M5**	**Fund**
Year 1	240	430	300	400	500	1000
Year 2	140	430	350	300	400	1100
Year 3	0	410	300	200	450	1200
Return	800	1600	1200	1480	1900	-

Table Q4(a)

The available funds for the various years are shown in the last column of the table. Also shown in the table are the returns for each model. The company can take on a maximum of three projects. At any time, it must engage in a minimum of two projects. The company has to decide which models to produce over the next three-year period. Formulate the linear programming problem above as a 0-1 variable model where the objective is to maximize profit.

(7 marks)

(b) Consider the following integer linear programming model.

$$\begin{aligned} \text{Maximize } \ z &= 36x_1 + 81x_2 \\ 3x_1 + 2x_2 &\leq 18 \\ 6x_1 + 15x_2 &\leq 75 \\ x_1, x_2 &\geq 0 \\ x_1, x_2 &\text{ are integers.} \end{aligned}$$

(i) Determine graphically, the solution to the continuous problem.

(5 marks)

(ii) Employ the branch and bound method to solve the integer linear programming model. Draw the tree diagram for the solution process. Construct a table of results using the following format.

Problem, I_k	**Problem solved, C_k**	**Solution to C_k**	**Current bound**	**Problem to pursue**	**Stored problem**

Table Q4(b)(ii)

(13 marks)

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